

CS 4032 – Web Programming



HTML – HyperText Markup Language



What is HTML ?

- The publishing language of the World Wide Web
- WWW – is a system of interlinked hypertext documents accessed via the Internet
- A mark-up language (i.e. its elements, HTML tags, “mark” text, instruct interpreter how to display text and also, “hyper-links” one document to another)
- Created by Tim Berners-Lee at CERN (1990)
- HTML specification is standardized and maintained by World Wide Web Consortium (w3.org)



Elements of HTML

- The elements are called tags
- The general form of a tag is:
- `<tag-name attribute="val" ... attribute="val" event="..function..">`
 - ... text content ...
 - `</tag-name>`
- Tags tell the browser how the “content text” should be displayed in the web page
- Attribute specifies characteristics of the tags
- Event (you will see later, in the JavaScript)
- Tag names are not case sensitive
- `<!-- ... -->` defines a comment

First, “Hello world”, HTML document

```
<!doctype html>  
<html>  
  <head>  
    <title>Hello world page</title>  
  </head>  
  <body>  
    Hello World!  
  </body>  
</html>
```

Main HTML tags

- An HTML document has a hierarchical structure
- `<html>` - is the root of the html document
- `<head>` - contains metadata about the document, action-scripting (see JavaScript), styles (see CSS) and general information referenced in the whole document
- `<body>` - contains the actual text of the document

Global attributes

- Core attributes:
 - class – specifies a class name for the element (CSS)
 - id – specifies an ID for the element
 - style – specifies an inline style for the element (CSS)
 - title – specifies extra information for the element (tooltip text)
 - content editable – specifies if content is editable(true/false)
 - draggable – if element can be dragged and dropped (true/false)
 - hidden – if element is hidden

https://www.w3schools.com/tags/ref_standardattributes.asp
- Language attributes:
 - dir="ltr | rtl" – specifies text direction for the contents of the element
 - lang – specifies a language code for the contents of the element
 - translate – specify whether content should be translated if page is localized
- Keyboard attributes:
 - accesskey – specifies a keyboard shortcut to access the element.
 - tabindex – specifies the tab order of the element

Type of tags

- Metadata tags
- Section tags
- Text-level appearance tags
- Grouping tags
- Image tag
- Anchor tag
- Table tag
- Script tag
- Embedded content tags

Metadata tags

- <title> - provides a title for the document

Ex.: <head>

<title>My title</title>

<head>

...

- <base> - specifies a default URL (Uniform Resource Locator) and a default target for all the links on a page

<base href="..URL.." target="_blank | _parent | _self | _top |
frameName">

Ex.: <head>

<base href=<http://www.cs.xyz.edu.pk/~forest>>

<base target="_blank">

</head>

Metadata tags (2)

- `<link>` - defines the relationship between a document and an external resource; appears in the head section, any number of times
- Attributes:
 - href - location of the linked document
 - rel – relationship between current document and linked document: alternate, appendix, bookmark, chapter, contents, copyright, glossary, help, home, index, next, prev, section, start, stylesheet, subsection
 - rev - relationship between linked document and current document; values the same as above
 - type – MIME type of the linked document
 - target – where the document is to be loaded
- Ex.: for including a stylesheet
- `<link rel="stylesheet" type="text/css" href="theme.css">`

Metadata tags (3)

- `<meta>` - describes information about the html document; it is not displayed, it is machine readable; appears in the head section
Ex.: `<meta name="description" content="Simple html page">`
`<meta name="author" content="Adrian Sterca">`
`<meta name="keywords" content="html, www">`
`<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">`
- `<style>` - defines style information in an html document (CSS)
Ex.: `<style type="text/css">`
`h1 {color:red}`
`p {color:blue}`
`</style>`

Section tags

- `<body>` - defines the document body which contains all the text, hyperlinks, tables, images etc.

Deprecated attributes: (use stylesheets instead)

alink – color of an active link; `rgb(x,x,x)`, `#xxxxxx`, `colorname`

bgcolor – background color; values as above

link – default color for unvisited link; values as above

text – color of the text; values as above

vlink – color of visited links; values as above

background = URL – background image

- `<head>` - defines the head section of the document; can contain tags: `<base>`, `<link>`, `<meta>`, `<script>`, `<style>`, `<title>` (required)

Section tags (2)

- `<div>` - defines a section in an HTML file; groups together elements which will be formatted using the same style (in HTML5 has no special meaning at all)
- `<frameset>` - defines a set of frames; mutually exclusive with `<body>`

Attributes:

cols = pixels | % | * - the number and size of columns in a frameset

rows = pixels | % | * - the number and size of rows in a frameset

Ex.: `<frameset cols="25%,*,25%">`

`<frame src="frame_a.htm">`

`<frame src="frame_b.htm">`

`<frame src="frame_c.htm">`

`</frameset>`

- `<frame>` and `<iframe>` - defines frames (windows) within a frameset or in a document

Section tags (3)

- `<frame>` - a frame (window) within a frameset Attributes:
 - `frameborder = 0 | 1`
 - `marginheight = pixels` (top and bottom margins of a frame)
 - `marginwidth = pixels` (left and right margins of a frame)
 - `name` – name of a frame
 - `noresize = "noresize"`
 - `scrolling = "yes | no | auto"`
 - `src = "URL"` - URL of the document to show in a frame
- `<iframe>` - an inline frame that contains another document within the current document
 - Attributes: `align`, `frameborder`, `height`, `width`, `marginwidth`, `marginheight`, `name`, `scrolling`, `src`

Text-level appearance tags

- `<h1>..

- defines headings <h2>This is a heading2</h2> search engines use headings to index the structure of your web page browsers automatically add an empty line before and after a heading`
- `<p>` - defines a paragraph
`<p>This is a paragraph</p>`
html documents are divided into paragraphs
browsers automatically add an empty line before and after a paragraph

Text-level appearance tags

- `<b>` - bold
- `<i>` - italic
- `<em>` - emphasized text
- `<strong>` - strong text
- `<s>` - strikethrough text
- `<u>` - underlined text
- `<del>` - deleted text
- `<sub>` and `<sup>` - subscript and superscript text
- `<pre>` - preformatted text (text is displayed exactly as is written)
- `<small>` - small text
- `<big>` - big text

Grouping tags

- Definition lists: <dl>, <dd>, <dt>
- Ex.:

<dl>

<dt>Name1</dt>

<dd>Name1 is something1</dd>

<dt>Name2</dt>

<dd>Name2 is something2</dd>

<dt>Name3</dt>

<dd>Name3 is something3</dd>

</dl>

Name1

Name1 is something1

Name2

Name2 is something2

Name3

Name3 is something3

Grouping tags (2)

- Ordered list: ,
- Ex.:

Ferrari

Audi

BMW

Ford

1. Ferrari
2. Audi
3. BMW
4. Ford

Grouping tags (3)

- Unordered list: ,
- Ex.:

Ferrari

Audi

BMW

Ford

- Ferrari
- Audi
- BMW
- Ford

Grouping tags (4)

- Drop-down list: <select>, <option>
- Ex.:

<select>

<option value="ferrari">Ferrari</option>

<option value="audi">Audi</option>

<option value="bmw">BMW</option>

<option value="ford">Ford</option>

</select>



Image tag

- `<img>` - embeds an image into html document
- Ex.: ``
`<img src=http://www.google.com/t.jpg>`
- Required attributes: src, alt
- Optional attributes:
 - align = top | bottom | middle | left | right
 - border = pixels
 - height = pixels | %
 - width = pixels | %
 - hspace = pixels
 - vspace = pixels

Anchor tag

- <a> - links current document to another document or a section from the current document
 - Ex.:
 - 1) <a href=<http://www.google.com>>google
 - 2) some text
-
- 1) – for linking this html document to an external document
(e.g. www.google.com)
 - 2) – for creating a bookmark inside the document; it is not displayed by the browser, it is invisible; this bookmark can be referenced by:
URL_of_this_document#test
ex: <http://www.google.com/index.html#test>

Anchor tag (2)

- Attributes:
 - href: the URL of the destination of the link
 - name: the name of the anchor (bookmark)
 - rel: relationship between the current document and the linked document
 - rev: relationship between the linked document and the current document
 - shape: the shape of the link in image map (default, rect, circle, poly)
 - target: where to open the linked document (_blank, _parent, _self, _top, framename)

Table tag

- `<table>` - defines a table
- `<th>` - defines a table header cell (bold and centered)
- `<tr>` - defines a table row
- `<td>` - defines a table data
- Attributes:
 - for `<table>`: border (pixels), cellpadding (pixels), cellspacing (pixels), frame, rules, summary (text), width (pixels,%)
 - for `<th>` and `<td>`: abbr (text), align (left, right, center, justify, char), axis, bgcolor - deprecated, char (character), charoff (number), colspan (number), headers, rowspan (number), height – deprecated, width – deprecated, scope (row, rowgroup, col, colgroup), valign (top, middle, bottom, baseline)
 - for `<tr>`: align, bgcolor - deprecated, char, charoff, valign

Table tag (2)

```
<table border="1">
<tr>
  <th>Professor</th>
  <th>Course</th>
  <th>Year of study</th>
</tr>
<tr>
  <td>John Boyd</td>
  <td>Operating Systems</td>
  <td>2</td>
</tr>
<tr>
  <td rowspan="2">Frank Black</td>
  <td>Web Programming</td>
  <td>3</td>
</tr>
<tr>
  <td>Computer Networks</td>
  <td>3</td>
</tr>
<tr>
  <td>Jack O'Neil</td>
  <td>Satellite Communications</td>
  <td>3</td>
</tr>
</table>
```

Professor	Course	Year of study
John Boyd	Operating Systems	2
Frank Black	Web Programming	3
	Computer Networks	3
Jack O'Neil	Satellite Communications	3

Script tag

- For inserting action scripting (JavaScript)
- Ex.:

```
<script type="text/javascript">  
    ... javascript code ...  
</script>
```
- Attributes
 - src: URL of the script
 - defer: execution of the script should be delayed until the page has loaded
 - charset: specifies encoding used in an external file

Embedded content tags

- <object> - including objects like images, audio, video, Java applets, ActiveX, pdf, flash
- Ex.:
- ```
<object classid="clsid:F08DF954-8592-11D1-B16A-00C0F0283628"
id="Slider1" width="100" height="50">
 <param name="BorderStyle" value="1" />
 <param name="MousePointer" value="0" />
 <param name="Enabled" value="1" />
 <param name="Min" value="0" />
 <param name="Max" value="10" />
</object>
```

# Other tags

- `<br>` - moves to the next line
- `<hr>` - draws a horizontal row

# New structural tags in HTML 5

- HTML 5 – focus on semantics of elements
- `<div>` should be used as the last resort, if no other structural element is suitable

- `<main>`
- `<section>`
- `<article>`
- `<footer>`
- `<header>`
- `<aside>`
- `<nav>`



# <main> tag

- Represents the main content of the <body> of the document
- Includes content that is unique to that document and excludes data that is repeated across a set of documents (e.g., header, footer, navigational links, copyright)
- Is no sectioning element
- There should be one visible main element in the document

# <section> tag

- Represents a generic section of a document
- Groups thematic content
- Ex:

<section>

<p>Web programming studies html, css and js</p>

<p>Tim Berners-Lee invented the www</p>

</section>

<section>

<p>Operating systems are essential software for computers</p>

<p>OS example: Unix, Linux, MacOS</p>

</section>

# <article> tag

- Represents complete, or self-contained, composition in a document
- Ex.:

<article>

HTML 5 was standardized in late 2014. It includes new structural-semantic elements like 'article' and 'section'

...

</article>



# <aside> tag

- Represents content that is not in the main flow of the document

# <footer>, <header> tags

<header>

Welcome to the page of A. Sterca.

</header>

<footer>

<a href="..">Back to index...</a>

</footer>

# <nav> tag

- A section with navigational links.

```
<nav>
```

```
Home
```

```
Users
```

```
Actions
```

```
</nav>
```

# <svg>,<audio>,<video> tags

- `<svg width="200" height="250">`  
    `<rect x="10" y="10" width="30" height="30"`  
    `stroke="black" fill="transparent" stroke-width="5"/>`  
    `</svg>`
- `<video width="320" height="240" controls>`  
    `<source src="movie.mp4" type="video/mp4">`  
    `<source src="movie.ogg" type="video/ogg">`  
    `</video>`

# <figure>,<figurecaption> tags

<figure>



<figcaption>Bubbles at work</figcaption>

</figure>

Thank you!



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